Trustworthy Machine Learning

Conformal Prediction

Sangdon Park

POSTECH

Conformal Prediction

Vladimir Vovk

FOLLOW

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Machine Learning Conformal Prediction Foundations of Probability Foundations of Statistics
Mathematical Finance

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Algorithmic learning in a random world V Vovk, A Gammerman, G Shafer Springer	1435	2005
Ridge regression learning algorithm in dual variables C Saunders, A Gammerman, V Vovk	1044	1998
Aggregating strategies V Vovk Proceedings of 3rd Annu. Workshop on Comput. Learning Theory, 371-383	956	1990
A Tutorial on Conformal Prediction. G Shafer, V Vovk Journal of Machine Learning Research 9 (3)	853	2008

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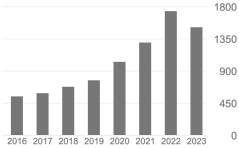


Conformal Prediction

Vladimir Vovk

Algorithmic learning in a random world

Authors	Vladimir Vovk, Alexander Gammerman, Glenn Shafer
Publication date	2005/3/1
Volume	29
Publisher	Springer
Description	This book is about conformal prediction, an approach to prediction that originated in machine learning in the late 1990s. The main feature of conformal prediction is the principled treatment of the reliability of predictions. The prediction algorithms described—conformal predictors—are provably valid in the sense that they evaluate the reliability of their own predictions in a way that is neither over-pessimistic nor over-optimistic (the latter being especially dangerous). The approach is still flexible enough to incorporate most of the existing powerful methods of machine learning. The book covers both key conformal predictors and the mathematical analysis of their properties. Algorithmic Learning in a Random World contains, in addition to proofs of validity, results about the efficiency of conformal predictors. The only assumption required for validity is that of" randomness"(the prediction algorithm is presented with …
Total citations	Cited by 1435

Scholar articles	Algorithmic learning in a random world V Vovk, A Gammerman, G Shafer - 2005 Cited by 1435 Related articles All 10 versions

> ...we are *hedging* the prediction — we are adding to it a statement about how strongly we believe it.
> – Vovk et al., 2005

Motivation

Conventional prediction:

$\hat{y}\Big($ $\Big) = \widehat{\text{Bulldog}}$

Conformal prediction:

$\hat{C}\Big($ $\Big) = \left\{ \begin{array}{c} \text{Toy terrier} \\ \text{Bulldog} \\ \text{poodle} \end{array} \right\}$

Motivation

Conventional prediction:

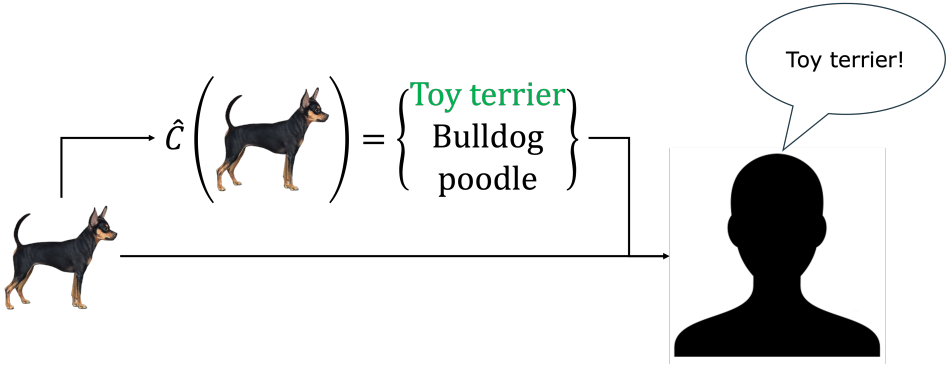
$$f : \mathcal{X} \mapsto \mathcal{Y}$$

Conformal prediction:

$$C : \mathcal{X} \mapsto 2^{\mathcal{Y}}$$

- Conventional prediction is a "point" prediction.
- Conformal prediction is a set-valued prediction.
- The set contains "likely-correct" alternative options.
  - The set size measures "uncertainty"!
- Why not confidence prediction? User-friendly?

Motivation: Decision Making



Why "Conformal"?

mathoverflow

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PUBLIC

How is the "conformal prediction" conformal?

Asked 6 years, 5 months ago Modified 6 months ago Viewed 2k times

16

+100

Thanks for your interest. The term "conformal prediction" was suggested by Glenn Shafer, and at first I did not like it exactly for the reason that you mention: it has nothing (or very little) to do with conformal mappings in complex analysis. But then I discovered other meanings, even in maths; e.g., Wikipedia has five on its disambiguation page for "conformal":

- Conformal film on a surface (same thickness)
- Conformal fuel tanks on military aircraft
- Conformal coating in electronics
- Conformal hypergraph, in mathematics
- Conformal software, in ASIC Software

So the word did not look taken to me anymore. The expression that we had used before Glenn proposed "conformal prediction" was even worse ("transductive confidence machine").

Thanks to Hengrui Luo for drawing my attention to this question.

As for question (2), the answer depends on which robust predictors you have in mind. The predictors with most similar properties are the ones in classical statistics (such as the standard prediction intervals in linear regression based on Student's t distribution); the main difference is that they are parametric. There is a predictive version of tolerance intervals in nonparametric statistics, but their treatment of objects (x parts of observations (x,y), where y are labels) is limited. Upper bounds on the probability of error given by standard PAC predictors are often too high to be useful.

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answered Apr 13, 2017 at 7:35

Vladimir Vovk

Conformal (Prediction) Sets

Definition (conformal set)

$$C(x) := \{y \in \mathcal{Y} \mid f(x, y) \geq q\}$$

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- We are using more recent notations based on inductive conformal prediction.
  - The notations are from Lei et al. [2018], Vovk et al. [2005], Tibshirani et al. [2019], and their combination.
  - Note that inductive conformal prediction [Papadopoulos et al., 2002] is an efficient variation of full conformal prediction [Vovk et al., 2005].

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  - Note that inductive conformal prediction [Papadopoulos et al., 2002] is an efficient variation of full conformal prediction [Vovk et al., 2005].
- $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$: a conformity scoring function
  - Measures how well (x, y) *conforms* to a trained model f (via a proper training set)
  - $f(x, y)$ is a likelihood of x for being y

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  - $f(x, y)$ is a likelihood of x for being y
- q: A parameter to be chosen by an algorithm.

Conformity Scoring Functions I

Conformity scoring functions can be (almost) any model!

Example (classification)

$$f(x, y) \coloneqq f_{\mathsf{cls}}(x, y)$$

- f_{cls}: a classification model, *e.g.*, resnet

Conformity Scoring Functions II

Conformity scoring functions can be (almost) any model!

Example (standard regression in 1-dimension)

$$f(x, y) \coloneqq -|\mu(x) - y|$$

- μ: a regressor

Conformity Scoring Functions III

Conformity scoring functions can be (almost) any model!

Example (probabilistic regression)

$$f(x, y) := \mathcal{N}(y; \mu(x), \sigma^2_{1:d}(x))$$

- *i.e.*, a Gaussian model with a diagonal covariance matrix [Nix and Weigend, 1994])
- d: The dimension of $\mathcal{Y}$.
- Implementation: $\mu(x) = f_{\text{mu}}(x)$ and $\ln \sigma^2 = f_{\text{var}}(x)$
  - $f_{\text{mu}}(x)$: a neural network
  - $f_{\text{var}}(x)$: a neural network

Back to Conformal Sets

Definition (conformal sets)

$$C(x) \coloneqq \{y \in \mathcal{Y} \mid f(x, y) \geq q\}$$

- A conformity scoring function f is given.
- f is a target to measure uncertainty.
- How to choose q?

Assumption: Exchangeability

Assumption

A sequence of random variables $X_1, X_2, \ldots$ is exchangeable if for any permutation σ, the following holds:

$$\mathbb{P}\left\{X_1 = x_1, X_2 = x_2, \ldots\right\} = \mathbb{P}\left\{X_{\sigma(1)} = x_1, X_{\sigma(2)} = x_2, \ldots\right\}.$$

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- The i.i.d. assumption implies the exchangeability assumption (why?).

A Goodness Metric: Coverage Guarantee

Definition (coverage guarantee)

$$\mathbb{P}\left\{Y_{n+1} \in \hat{C}(X_{n+1})\right\} \geq 1 - \alpha$$

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- $\hat{C}$: A conformal set constructed by the training set

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- (X_i, Y_i) for $i = 1, \ldots, n+1$: the exchangeable samples (thus the i.i.d. samples)
- $\hat{C}$: A conformal set constructed by the training set
- $1 - \alpha \in (0, 1)$: A desired coverage rate

Quantile

Quantile of a Distribution

The level β quantile of a distribution F:

Definition (quantile)

$$\text{Quantile}(\beta; F) := \inf\{z \mid \mathbb{P}\{Z \leq z\} \geq \beta\}$$

- F: a distribution over the augmented real line, $\mathbb{R} \cup \{\infty\}$
- $Z \sim F$
  - allows multiple instances of the same element

Quantile

Quantile of an Empirical Distribution

The level β quatile of an empirical distribution of the values $v_{1:n}$:

Definition (quantile)

$$\text{Quantile}(\beta; v_{1:n}) \coloneqq \text{Quantile}\left(\beta; \frac{1}{n}\sum_{i=1}^{n}\delta_{v_i}\right)$$

- $v_{1:n} \coloneqq \{v_1, \ldots, v_n\}$: an unordered multiset
- δ_a: a δ-distribution (*i.e.*, a point mass at a)

Quantile Lemma

Lemma (Tibshirani et al. [2019])

If $V_1, \ldots, V_{n+1}$ are exchangeable random variables, then for any $\beta \in (0,1)$, we have

$$\mathbb{P}\left\{V_{n+1} \leq \textit{Quantile}(\beta; V_{1:n} \cup \{\infty\})\right\} \geq \beta.$$

- The key lemma for conformal predictionx

Quantile Lemma: A Proof Sketch I

- One fact about quantiles of a discrete distribution F with support points $a_1, \ldots, a_k \in \mathbb{R}$:
  - Denote $q := \text{Quantile}(\beta; F)$
  - Reassign the points a_i strictly larger than q to arbitrary values also strictly larger than q, yielding a new distribution $\tilde{F}$
  - Still we have $\text{Quantile}(\beta; F) = \text{Quantile}(\beta; \tilde{F})$
- Thus, we have

$$V_{n+1} > \text{Quantile}(\beta; V_{1:n} \cup \{\infty\}) \iff V_{n+1} > \text{Quantile}(\beta; V_{1:n+1}).$$

- This implies

$$\begin{aligned} \mathbb{P}\Big\{V_{n+1} \leq \text{Quantile}(\beta; V_{1:n} \cup \{\infty\})\Big\} &= \mathbb{P}\Big\{V_{n+1} \leq \text{Quantile}(\beta; V_{1:n+1})\Big\} \\ &\geq \frac{\lceil \beta(n+1) \rceil}{n+1} \\ &\geq \beta. \end{aligned} \tag{1}$$

Quantile Lemma: A Proof Sketch II

- Why (1)?
  - By exchangeability, we have for any integer $k \in \{1, \ldots, n+1\}$,

$$\mathbb{P}\Big\{V_{n+1} \leq V_{[k]}\Big\} \geq \frac{k}{n+1},$$

where $[k]$ is the k-th smallest value of $V_1, \ldots, V_{n+1}$.

Quantile Lemma: A Proof Sketch III

- Suppose that there is no tie (see Kuchibhotla [2020] for a general proof). We have

$$
\begin{aligned}
\mathbb{P}\Big\{V_{n+1} \leq V_{[k]}\Big\} &\geq \mathbb{P}\left\{\bigvee_{i=1}^{k} V_{n+1} = V_{[i]}\right\} \\
&= \sum_{i=1}^{k} \mathbb{P}\Big\{V_{n+1} = V_{[i]}\Big\} \\
&= \sum_{i=1}^{k} \frac{n!}{(n+1)!} \\
&= \frac{k}{n+1}.
\end{aligned}
\tag{2}
$$

Quantile Lemma: A Proof Sketch IV

- Why (2)?
  - For each permutation π, we have

$$\begin{aligned}\mathbb{P}\{V_1 \leq \cdots \leq V_{n+1}\} &= \mathbb{P}\{(V_1, \ldots, V_{n+1}) \in A\} \\ &= \mathbb{P}\{(V_{\pi(1)}, \ldots, V_{\pi(n+1)}) \in A\} \\ &= \mathbb{P}\{V_{\pi(1)} \leq \cdots \leq V_{\pi(n+1)}\}\end{aligned} \tag{3}$$

    where $A := \{(x_1, \ldots, x_{n+1}) \mid x_1 \leq \cdots \leq x_{n+1}\}$ and (3) holds due to the exchangeability assumption.
  - This means that "exchangeability" implies "uniform probability over orders"

Quantile Algorithm

Definition (quantile algorithm)

Given $(X_1, Y_1), \ldots, (X_n, Y_n)$,

$$\hat{q}_{1-\alpha} \coloneqq \text{Quantile}(1-\alpha, V_{1:n} \cup \{\infty\}),$$

where $V_i \coloneqq -f(X_i, Y_i)$.

- The implementation is as simple as finding the k-th smallest value.

Coverage Guarantee of the Quantile Algorithm

Theorem (Vovk et al. [2005], Lei et al. [2018])

Assume that (X_i, Y_i) *for* $i \in \{1, \ldots, n+1\}$ *are exchangeable. For any scoring function* f *and any* $\alpha \in (0, 1)$*, denote the conformal set by*

$$\hat{C}(x) := \left\{ y \in \mathcal{Y} \;\middle|\; -f(x, y) \leq \hat{q}_{1-\alpha} \right\}.$$

Then, we have

$$\mathbb{P}\left\{ Y_{n+1} \in \hat{C}(X_{n+1}) \right\} \geq 1 - \alpha,$$

where the probability is taken over (X_i, Y_i)*.*

- This is a marginal coverage guarantee.

Coverage Guarantee of the Quantile Algorithm: A Proof Sketch

- Observe that

$$Y_{n+1} \in \hat{C}(X_{n+1}) \iff V_{n+1} \leq \text{Quantile}(1-\alpha, V_{1:n} \cup \{\infty\})$$

- Due to the symmetric construction of scores (using the same scoring function f), for any permutation π we have

$$(Z_1, \ldots, Z_{n+1}) \stackrel{d}{=} (Z_{\pi(1)}, \ldots, Z_{\pi(n+1)}) \implies (V_1, \ldots, V_{n+1}) \stackrel{d}{=} (V_{\pi(1)}, \ldots, V_{\pi(n+1)})$$

  where $Z_i := (X_i, Y_i)$.
- As $(Z_1, \ldots, Z_{n+1})$ are exchangeable, so are $(V_1, \ldots, V_{n+1})$.
- Use the quantile lemma.

Power of Conformal Prediction

The coverage guarantee is drawn with minimal assumptions.

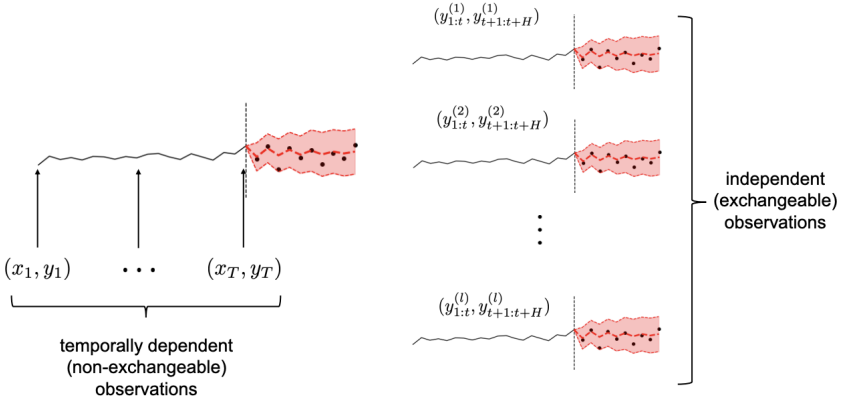
- It does not make assumptions on a distribution except for the exchangeability.
- The guarantee holds for any conformity scoring function.

Size of Conformal Sets

- Application-dependent issues
  - classification: set size
  - 1-D regression: interval length
  - multi-dimentional regression: *e.g.*, volume
- Larger set: uncertain (*e.g.*, the entire set)
- Smaller set: more certain (*e.g.*, a singleton)
- We will see some analysis in PAC conformal prediction.

Interesting Variation: Time-series Forecasting

Conformal Time-series Forecasting [Stankeviciute et al., 2021]



- Conformal prediction for independent time-series data
  - *e.g.*, temperature change for each year

Conformal Time-series Forecasting I

Problem

Setup:

- $y_{t:t'} := (y_t, y_{t+1}, \ldots, y_{t'}) \in \mathbb{R}^d \times \ldots \mathbb{R}^d$: a time-series of d-dimensional observation
  - Let $d = 1$
- H: a prediction horizon
- $\hat{y}_{t'+1:t'+H}$: predicted future observations (*e.g.*, the output of a RNN)
- $C_{t+h}(y_{1:t})$: a prediction interval at time $t + h$
  - $C_{t+h}(y_{1:t}) := [\hat{y}^L_{t+h}, \hat{y}^U_{t+h}]$

Conformal Time-series Forecasting II

Problem

Desired coverage guarantee:

$$\mathbb{P}\Big\{\forall h \in \{1, \ldots, H\},\ y_{t+h} \in C_{t+h}(y_{1:t})\Big\} \geq 1 - \alpha$$

- The probability is taken over $y_{1:t+H}$.
- $1 - \alpha$: a desired coverage rate

Goal: Find C_{t+h} for all $h \in \{1, \ldots, H\}$.

Conformal Time-series Forecasting

Approach

- $\mathcal{D} := \{(y_{1:T}^{(i)}, y_{T+1:T+H}^{(i)})\}_{i=1}^{m}$: a calibration set

Conformal Time-series Forecasting

Approach

- $\mathcal{D} := \{(y^{(i)}_{1:T}, y^{(i)}_{T+1:T+H})\}_{i=1}^{m}$: a calibration set
- Observe that

$$\mathbb{P}\Big\{\exists h \in \{1, \ldots, H\},\ y_{t+h} \notin C_{t+h}(y_{1:t})\Big\} \leq \sum_{h \in \{1,\ldots,H\}} \mathbb{P}\{y_{t+h} \notin C_{t+h}(y_{1:t})\} \tag{4}$$

$$\leq \alpha \tag{5}$$

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  - (4) holds due to the union bound.

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  - (5) holds if $\mathbb{P}\{y_{t+h} \notin C_{t+h}(y_{1:t})\} \leq \frac{\alpha}{H}$

Conformal Time-series Forecasting

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$$\leq \alpha \tag{5}$$

  - (4) holds due to the union bound.
  - (5) holds if $\mathbb{P}\{y_{t+h} \notin C_{t+h}(y_{1:t})\} \leq \frac{\alpha}{H}$
- Due to the standard conformal prediction, we can find C_{t+h} such that

$$\mathbb{P}\{y_{t+h} \notin C_{t+h}(y_{1:t})\} \leq \frac{\alpha}{H}.$$

Conclusion

- Conformal prediction is a powerful tool to construct a prediction set (for measuring uncertainty) with correctness guarantees.
- Conformal prediction has many applications due to its "distribution-free" and "scoring-function-free" nature.
- The original conformal prediction framework can be extended to "conditional" cases (*e.g.*, PAC conformal prediction).

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